

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Linear Algebra	Course Code:	MT-1004
Program:	BDS	Semester:	Spring 2026
Paper Date:	11-05-26		
Section:			
Exam:	Work Sheet (Chapter 8 and 4.7)		

Name :

Roll No:

Question 1: Let $T : P_2 \rightarrow P_2$ be the contraction operator with factor $k = 1/4$.

- (a) Find $T(1 + 4x + 8x^2)$.
- (b) Find the rank and nullity of T .

Question 2: Compute $(T_3 \circ T_2 \circ T_1)(x, y)$.

Where

$$T_1(x, y) = (-2y, 3x, x - 2y),$$

$$T_2(x, y, z) = (y, z, x),$$

$$T_3(x, y, z) = (x + z, y - z)$$

Question 3: Determine whether the matrix transformation $T_A: R^3 \rightarrow R^3$ is an isomorphism.

i) $A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$

ii) $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$

Question 4: Ex . 8.4 Q1

Question 5: Ex. 4.7 Q3